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# Allogeneic Bone Marrow Transplantation in Conventional Mice: I. Effect of Antibiotic Therapy on Long Term Survival of Allogeneic Chimeras

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**Summary.** In the present communication the beneficial effect of long term antimicrobial treatment with poorly absorbable antiboitics on the survival of allogeneic bone marrow chimeras was investigated.

The combination of $C^{57}Bl$ mice as bone marrow donors and CBA/CA mice as irradiated recipients (800 rad) was used because of their strong histoincompatibility on the H-2 loci. All allografted recipients received $10 \times 10^6$ bone marrow cells. The majority of the recipients, which were rendered gnotobiotic by an antimicrobial treatment, achieved stable long term chimerism. In contrast, the conventional chimeras died from secondary disease within 9 weeks after transplantation. As early as 14 days after allogeneic bone marrow grafting the gnotobiotic rceipients tolerated the reassociation with a conventional microflora without a change in the rate of mortality. Bone marrow cells ($8 \times 10^6$ i.v.) and spleen cells ($2 \times 10^6$ i.v.) collected from allogeneic chimeras failed to induce graft-versus-host-reaction (GVH) in a second lethally irradiated host.

The data indicate, that the high rate of mortality in murine allogeneic bone marrow chimeras results from delayed GVH-reaction and systemic infection. The marrow graft, once established seems to exert tolerance against the allogeneic host. The pathogenesis of the systemic infection has not yet been worked out. It is assumed that it originates from bacteremia, induced by radiation dependent lesions of the epithelial integrity and defected lymphatic tissue in the gut.

**Key words:** Allogeneic bone marrow transplantation – Antimicrobial treatment – Mitigation of secondary desease – Long term chimerism

**Zusammenfassung.** In der vorliegenden Arbeit wird an Hand eines tierexperimentellen Modells über allogene Knochenmarktransplantation die Bedeutung der gnotobiotischen Versuchsbedingungen für den Verlauf der Transplantat-gegen-Wirt (Graft-versus-host)-Reaktion (GVHR) untersucht.

Nach Konditionierung mit 800 rad Ganzkörperbestrahlung wurden in drei Experimenten CBA-CA-Mäuse mit $10 \times 10^6$ Knochenmarkzellen von $C^{57}Bl$-Mäusen transplantiert und die Überlebenszeit bestimmt.

Alle Empfängertiere, die keiner Antibiotikatherapie unterzogen wurden, starben in typischer Weise an den Folgen einer vom Transplantat ausgehenden Abstoßungsreaktion innerhalb von 9 Wochen. Diejenigen CBA/CA-Mäuse, die vor Bestrahlung und Transplantation mit schlecht resorbierbaren Antibiotika behandelt worden waren, überlebten zu einem Prozentsatz von 50% oder darüber. Sie entwickelten einen Langzeitchimärismus, der mittels Hämoglobinelektrophorese nachgewiesen wurde.

Die histologische Untersuchung der Organe, vor allem der Leber, bestätigte das Vorliegen einer GVHR in beiden Tiergruppen nach der Allotransplantation. Auffällig war die nahezu vollständige Rückbildung der für die Abstoßungsreaktion typischen Organveränderungen in den mit Antibiotika behandelten Chimären nach Ablauf von 9 Wochen.

Zur Charakterisierung der Bedeutung der Darmflora wurde eine Reassoziation der antibiotikatherapierten Chimären mit Keimen der normalen Darmflora vorgenommen. Es zeigte sich, daß bereits 14 Tage nach Transplantation eine Assoziation mit Keimen der speziesspezifischen Darmflora keine Änderung der Sterberate verursachte.

In einem weiteren Versuchsansatz wurden Knochenmark und Milz der Langzeitchimären auf ihre Fähigkeit hin untersucht, in einer zweiten Passage in einem allogenen Empfängertier (CBA/CA) eine GVHR zu erzeugen. Weder Knochenmarkzellen allein noch zusätzliche applizierte Milzzellen vermochten in einer Zweitpassage eine tödliche GVHR zu erzeugen.

Die vorliegenden Untersuchungen machen deutlich, daß es möglich ist, unter den beschriebenen Bedingungen einen Langzeitchimärismus nach allogener Transplantation zu erzeugen.

The successful management of secondary disease is still one of the major obstacles to the routine application of bone marrow transplantation in man. So far, the main strategies used have included the treatment of the grafted recipient with immunosuppressive drugs and manipulations of the donor cell suspensions. Recent studies indicate, that the degree of histoincompatiblity between the donor and host may not be the only ethiological factor affecting mortality in allogeneic bone marrow grafting. Van Bekkum [11] showed that antibiotic therapy during secondary disease improved the mean survival time of allogeneic chimeras. Jones et al. [6] reported on successful allogeneic bone marrow transplantation in germfree mice, which showed similar survival times as the isogeneic chimeras. However, when the mice were brought into contact with a conventional intestinal flora, 100% of the allogeneic chimeras died within 14 days. Similar results were obtained in a preliminary study of our own [3] in which conventional mice subjected to a regimen of long term antibiotic treatment were used as lethally irradiated allogeneic bone marrow recipients. In contrast to the data reported by Jones et al., however, all long term survivors of the allogeneic bone marrow chimeras could be successfully reassociated with an intestinal flora of locally reared conventional mice. None of the the animals developed symptoms of a relapse of secondary disease (SD). This observation was assured and analysed more in detail by Van Bekkum et al. [12].

It is the aim of this presentation to further investigate the effect of intensive antimicrobial treatment on the fate of conventional allografted bone marrow chimeras, kept in a gnotobiotic environment. Data will be presented which demonstrate the beneficial effect of long term antibiotic induced "decontamination" on the mortality of allogeneic bone marrow chimeras. Furthermore, the reduced capacity of chimeric bone marrow and spleen cells, to induce secondary disease related mortality, will be tentatively analysed.

## Materials and Methods

*Mice*

Female CBA/CA mice, 10–12 weeks old were used as bone marrow recipients. Female $C_{57}Bl$ mice 12–14 weeks old, served as bone marrow donors. This combination guarantees histoincompatibility in all genetic specificities of H-2 locus except for H-5 [9]. In addition, the difference in the hemoglobin type between both strains was used to study the establishment of hemopoietic chimerism.

*Irradiation*

Whole body irradiation was performed using a 250 kV 15 mA Stabilipan X-ray source with 1 mm Cu filtration. The distance between the focus of the X-ray tube and the bottom of the plastic cage containing the animals was 100 cm. All the CBA/CA mice were given 800 rad with a dose rate of 30 rad/min.

*Bone marrow transplantation*

Donor mice were killed by cervical dislocation. 20 femora were used to prepare a single cell suspension as described elsewhere [8].

An appropriate number of cells were suspended in Hank's BSS, and injected in 0.5 ml medium in a lateral tail vein, 24 h after whole body irradiation. $10 \times 10^6$ cells were injected for allogeneic bone marrow transplantation, where as isogeneic bone marrow recipients were grafted with $1 \times 10^6$ cells.

*Hb Electrophoresis*

In order to demonstrate hemopoietic chimerism, qualitative and quantitative hemoglobin electrophoresis was performed unsing the method of Kohne [7] and Dozy [2]. Both the isogeneic and the allogeneic animals were examined during the whole period of observation to assess the changes in the type of hemoglobin of peripheral blood erythrocytes. For this purpose and for investigations concerning the hemopoietic system 40 animals in each group (allogeneic and isogeneic recipients) were reserved and did not take part in the survival studies.

*Establishment of a gnotobiotic state*

a) Regimen of antibiotic treatment: One group of the allogeneic bone marrow recipients was treated with antibiotic drugs in order to achieve "decontamination" with respect to all microorganisms detectable by routine bacteriological methods. Treatment was started 14 days before transplantation. A combination of Neomycin, Carbenicillin and Bacitracin, dissolved in drinking water, was used in a final concentration of 4 g/l. Pimaricin (0.1 g/l) was added as a fungistatic drug. This therapy was given continuously throughout the whole period of observation.

b) Maintenance of a gnotobiotic environment: To prevent secondary contamination, all animals were kept in laminar flow benches, which were sterilized with peracetic acid before use. Benches with a nonturbulent horizontal airflow ("cross flow") were prefered, as they seem to be superior to protect from outside contamination. Sterile food enriched with vitamins, and drinking water was supplied ad libidum; the conventional controls which did not receive an antibiotic treatment were fed with the standard food offered to all mice kept in the local animal care unit. Bacteriological swabs of the feces and the bedding material were checked routinely, at least once a week, for aerobic and anaerobic microorganisms, using standard bacterilogical methods.

*Reassociation of the chimeras with members of the intestinal microflora 14 days after transplantation*
For reassociation the antibiotic treatment was stopped and after an interval of 7 days the chimeras were associated with aerobic and anaerobic members of the intestinal flora. $1 \times 10^3$ E. coli and $1 \times 10^3$ Streptococci faecalis as members of the aerobic flora were inoculted into the stomach through a tube. Three species of Clostridia and Propionibacterium acnes as members of the strict anaerobic flora were used alternatively. The mice were fed with fresh faecal pellets that consisted only of these microorganisms. Each associated animal group consisted of 15 decontaminated allogeneic bone marrow recipients. Four days after association the faecal samples of the chimeras were found to be positive for the flora inoculated.

## Results

### *Gnotobiotic state induction*

All CBA/CA included in the study had an intestinal flora found to be "normal" for conventional mice of our animal care unit, excluding pseudomonas species. The group of mice that was treated with antibiotics gave negative bacteriological swabs from the feces within 3–5 days application of the drugs. Five weeks later, all specimen from the skin and bedding material were negative. After 8 weeks of continuous antibiotic treatment, no positive cultures were found, even when the therapy was interrupted for a period of another 28 days.

### *Survival of conventional and "decontaminated" mice after bone marrow transplantation*

The bone marrow transplantation was performed after 2 weeks of antibiotic treatment and 24 h after 800 rad whole body irradiation.

Table 1 shows the results of one experiment. The percentage survival of the allogeneic conventional and decontaminated bone marrow recipients as well as that of the irradiated control mice was determined. All the CBA/CA mice that had not received the protective bone marrow transplant died within 20 days of receiving 800 rad. In these non-transplanted animals, no difference was found between the conventional mice and those treated with antibiotics. None of the mice grafted with allogeneic bone marrow died within that period. Thereafter increasing mortality was observed. None of the conventional animals lived longer than 70 days. In contrast, all the allogeneic bone marrow recipients that were treated with antibiotic drugs survived 30 days, and 70% were still alive after a further 40 days and 55% survived more than 120 days.

**Table 1.** Effect of antimicrobial treatment on the survival of lethally irradiated mice (CBA/CA) and allogeneic bone marrow recipients ($C_{57}Bl \rightarrow$ CBA/CA).
All bone marrow recipients were given $10.0 \times 10^6$ i.v., 24 h post irradiation

| Treatment post 800 rad | No. of Mice | %-Survival (Days) 10 | 20 | 30 | 40 | 50 | 60 | 70 | 120 |
|---|---|---|---|---|---|---|---|---|---|
| B.M.-Allograft and Antibiotes | 25 | 100 | 100 | 100 | 84 | 84 | 80 | 72 | 53 |
| B.M.-Allograft | 20 | 100 | 100 | 60 | 30 | 30 | 20 | 0 | — |
| Antibiotics | 10 | 100 | 0 | — | — | — | — | — | — |
| no Therapy | 10 | 100 | 0 | — | — | — | — | — | — |

Table 2 summarizes the results of a second experiment in which the survival of conventional isogeneic bone marrow recipients was compared to that of the decontaminated CBA/CA mice, which had been grafted with the allogeneic bone marrow of $C^{57}$Bl mice. 86% of the 50 isogeneic chimeras survived 168 days and 88% of 50 allogeneic bone marrow recipients survived 168 days.

**Table 2.** Long term survival of "decontaminated" allogeneic in comparison to isogeneic bone marrow recipients irradiated with 800 rad.
Whole body irradiation was performed 24 h prior bone marrow grafting

| B. Marrow-Donor | | Recipients | | %-Survival (Days) | | | | |
|---|---|---|---|---|---|---|---|---|
| Strain | Cell dose | Strain | No. | 10 | 20 | 40 | 80 | 168 |
| $C_{57}$Bl | $10.0 \times 10^6$ | CBA/CA | 50 | 100 | 100 | 88 | 88 | 88 |
| CBA/CA | $1.0 \times 10^6$ | CBA/CA | 50 | 100 | 86 | 86 | 86 | 86 |

*Evidence of chimerism*

By means of the hemoglobin electrophoresis all the transplanted CBA/CA mice were proved to be allogeneic bone marrow chimeras. First signs of donor type hemoglobin (Hb) in the allogeneic recipients could be detected between weeks 4 and 5; from week 7 onwards Hb specificities of the CBA/CA type had been completely exchanged for the characteristics of the $C_{57}$Bl mice (Table 3).

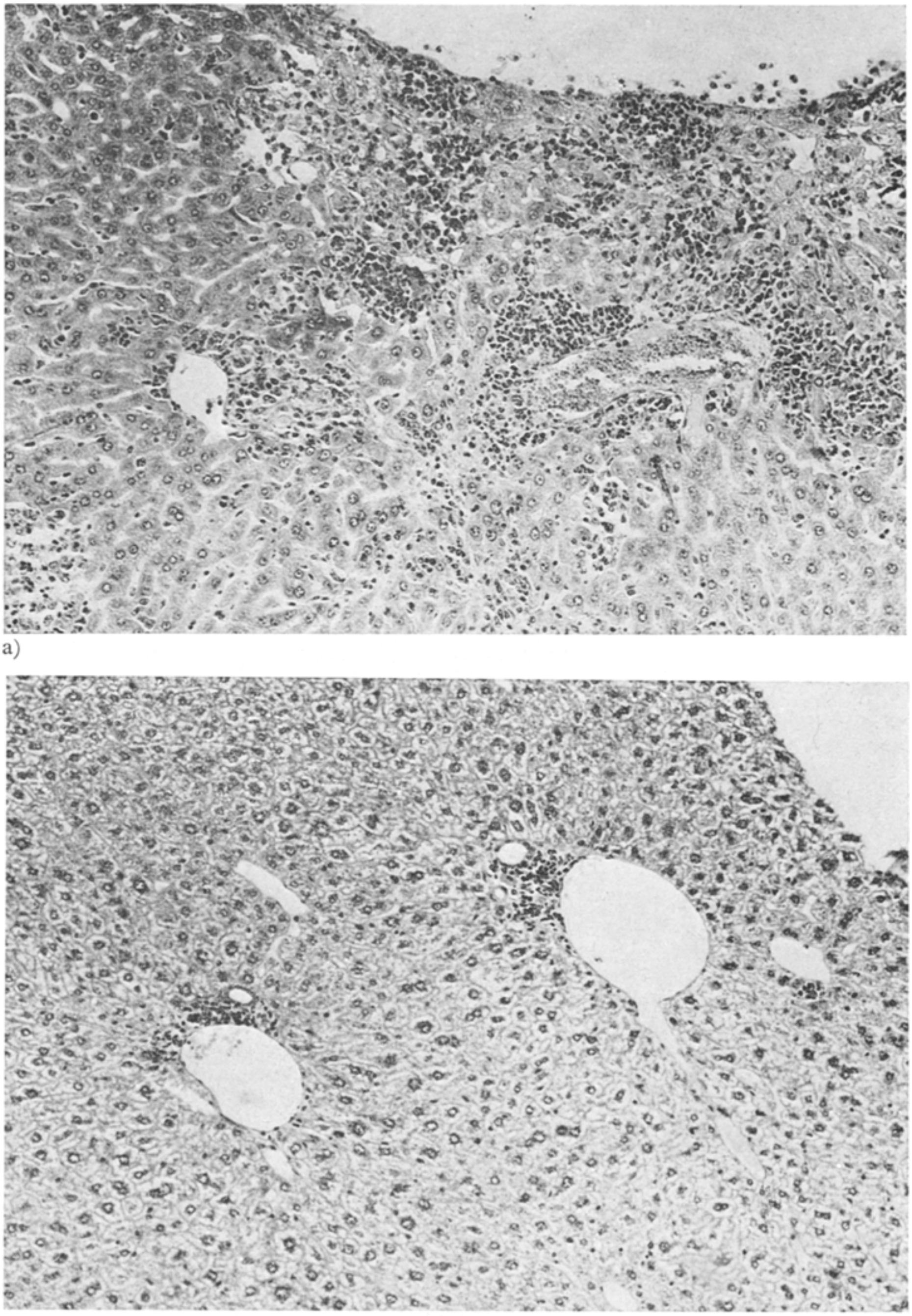
**Table 3.** Quantitative determination of Hb-components in allogeneic bone marrow chimeras ($C_{57}$Bl → CBA/CA)

| Source of Hemoglobin* (Hb) | Fraction I (%) | Fraction II (%) |
|---|---|---|
| CBA/CA (before grafting) | 14.20 | 85.80 |
| ($C_{57}$Bl → CBA/CA): | | |
| 2nd week | 9.17 | 90.83 |
| 3rd week | 7.50 | 92.50 |
| 4th week | 5.96 | 94.04 |
| 5th week | 4.69 | 95.31 |
| 6th week | 3.48 | 96.52 |
| 7th week | 1.30 | 98.70 |
| 8th week | — | 100.00 |
| 21th week | — | 100.00 |
| 38th week | — | 100.00 |
| CBA/CA (control group) | 15.03 | 84.97 |
| $C_{57}$Bl (control group) | — | 100.00 |

* Hb-Determination was performed with pooled blood of 3 animals (+ EDTA)

*Evidence of Graft versus Host Reaction (GVHR)*

Signs of secondary disease were found in all of the allogeneic bone marrow recipients between weeks 6 and 9 after transplantation. Diarrhea, loss of weight and hair as well as adynamia were observed during this time. All the signs, except the loss of hair, were less evident in the "decontaminated" group. In contrast to the



a)

b)

**Fig. 1.** Histomorphological picture of the GVH-reaction in the livers of
**a)** "conventional" and of
**b)** "decontaminated" allogeneic bone marrow recipients, 9 weeks after whole body irradiation and bone marrow grafting

antibiotic treated mice, the majority of the conventional allogeneic bone marrow recipients developed septicaemia and abscesses, predominantly in the lung, liver and spleen. The most common bacteriological isolate was Escherichia coli, followed by Proteus vulgaris. The animals, that survived the 9th week after bone marrow grafting recovered, and lost all symptoms of secondary disease.

Serial sections of the liver and the spleen were performed to document the organic manifestation of GVH-reaction. Focal necrosis and mononuclear cell infiltration were found to a similar extent in both animal groups from week 5 to 7 after bone marrow transplantation. No change in the histomorphological picture of the liver was observed in conventional chimeras (Fig. 1a) after further 2 weeks. In contrast, most of the organic lesions got repaired in the liver of the "decontaminated" animals by that time, except for perivascular infiltrations by mononuclear cell (Fig. 1b). After 17 weeks the livers and spleens of the "decontaminated" chimeras were free of signs of GVH-reaction.

*Early reassociation of allogeneic bone marrow chimeras with an intestinal flora*

To investigate the relevance of the microbial flora as an enhancing factor of GVH-reaction related mortality, decontaminated allogeneic chimeras were associated with members of a "conventional" intestinal flora, 14 days after bone marrow transplantation. Two groups were formed and the animals inoculated with members of the strict anearobic or aerobic intestinal flora. The intestinal take of each flora was assured by repeated bacteriological analysis of the feces. As shown in Figure 2, no difference in the rate of survival could be observed between both groups as compared to the decontaminated chimeras. After 10 weeks more than 50% were still alive. Furthermore, all chimeras showed comparable but mitigated signs of secondary disease between week 5 and 9 after bone marrow grafting.

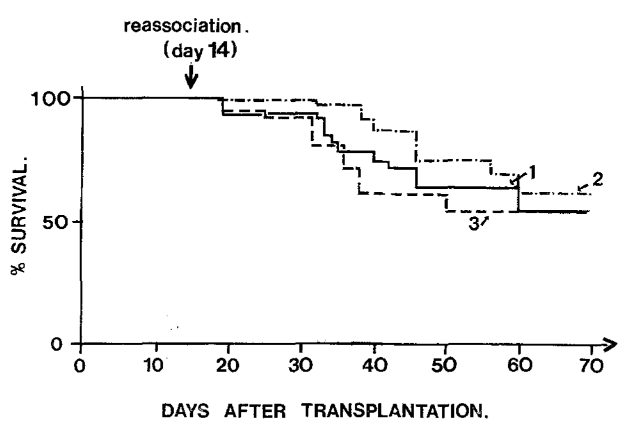



**Fig. 2.** Survival of "decontaminated" allogeneic bone marrow chimeras, reassociated with members of the "conventional" intestinal flora, 14 days after B. M. transplantation

1. "Decontaminated" chimeras (15 animals)
2. "Decontaminated" chimeras, associated with an aerobic flora (15 animals)
3. "Decontaminated" chimeras, associated with an anaerobic flora (15 animals)

*Retransplantation of chimeric bone marrow*

In order to investigate the ability of the chimeric bone marrow, to protect in a second passage radiation induced aplastic recipients, and furthermore, to induce GVHR in these animals the following experiment was performed: Bone marrow suspensions of allogeneic chimeras (40 weeks after transplantation) were grafted into "conventional", lethally irradiated CBA/CA or $C_{57}$Bl mice. To enhance the graft-versus-host-reaction in the CBA/Ca mice, which could be considered to be allogeneic to the primary origin of the donor cells, $2 \times 10^6$ spleen cells of the chimeras were added and injected together with the bone marrow cells intravenously. The results are summarized in Table 4. Compared to primary allografting the same bone marrow dose of $10 \times 10^6$ cells was found to be sufficient to protect lethally irradiated allogeneic recipients, while in the isogeneic situation $1 \times 10^6$ bone marrow cells were as effective to provide a comparable protection against postirradiation mortality. Again, hemopoietic chimerism was assured by Hb-electrophoresis. No signs of secondary disease were observed in any of the experimental groups (Table 4a). Even those chimeras, which were injected with both, bone marrow and spleen cell suspensions did not reveal increased mortality nor signs of secondary disease. In contrast, in conventional allogeneic recipients transplanted with identical doses of bone marrow and spleen cells of $C_{57}$Bl (= primary donor) the typical high percentage of mortality was observed (Table 4b).

**Table 4a.** Survial of lethally irradiated mice grafted with hemopoietic tissue of allogeneic bone marrow chimeras ($C_{57}$Bl → CBA/CA).
40 weeks after whole body irradiation and allogeneic bone marrow grafting

| Donor Cells | Cell Dose i. v. | Recipients (800 rad) Strain | No. | 100% Survival Days |
|---|---|---|---|---|
| B. Marrow | $1.0 \times 10^6$ | $C_{57}$Bl | 10 | 120 |
| B. Marrow | $10.0 \times 10^6$ | CBA/CA | 10 | 120 |
| B. Marrow + Spleen | $0.8 \times 10^6$ + $2.0 \times 10^6$ | $C_{57}$Bl | 10 | 120 |
| B. Marrow + Spleen | $8.0 \times 10^6$ + $2.0 \times 10^6$ | CBA/CA | 10 | 120 |

**Table 4b.** Survival of lethally irradiated mice grafted with hemopoietic tissue of "primary" allogeneic donors ($C_{57}$Bl) continuously exposed to antibiotic therapy (≙ "decontaminated")

| Donor Cells | Cell Dose i. v. | Recipients (800 rad) Strain | No. | 25% Survival Days |
|---|---|---|---|---|
| B. Marrow | $10.0 \times 10^6$ | CBA/CA | 20 | 32 |
| B. Marrow + Spleen | $10.0 \times 10^6$ $2.0 \times 10^6$ | CBA/CA | 20 | 17 |
| B. Marrow + Spleen | $10.0 \times 10^6$ $2.0 \times 10^6$ | CBA/CA* "dec" | 20 | 26 |

## Discussion

In agreement with earlier observations [3,6,12] the results presented herein demonstrate the crucial role of the gnotobiotic state to improve the chance of survival of allogeneic bone marrow grafting in mice. Mortality due to secondary disease may thus be understood as a result of both, the graft dependent immunological process directed against the allogeneic host and the appearance of bacterial infection.

In this context, it was of interest to study the effect of prolonged antibiotic treatment on recipients of allogeneic bone marrow. The regimen of anti-microbial therapy was found to eliminate all microorganisms, which could be detected by means of routine bacteriological methods. The treatment was understood to be supportive rather than sufficient for an analytical characterisation of the gnotobiotic state of the animals. However, the long-term survival of "decontaminated" allogeneic bone marrow chimeras which developed neither signs of focal nor systemic infection on the one hand, and the high mortality of conventional recipients due to GVH-reaction and systemic infections on the other, indicate, that the microflora was sensitive to the drug regimen used, including the species of microorganisms which seem to contribute to the lethal course of secondary disease.

It has been speculated about possible interactions between parts of the microflora and the allogeneic bone marrow graft resulting in the lethal course of GVH-reaction in "conventional" chimeras [12]. The successful early reassociation of "decontaminated" allochimeras (Fig. 2), however, indicates, that the presence of a "conventional" microflora, alone, does not simply imply an enhancement of the GVH-reaction.

Obviously the deleterious influence by the bacterial flora seems to gain its relevance in the very early phase after whole body irradiation and bone marrow transplantation. The mechanisms involved are still obscure. A radiation induced transient defect of the epithelial integrity and the concomintant cellular depletion of the lymphatic tissue in the gut may allow members of the intestinal flora to pass the mucosal barrier and circulate in the peripheral blood. During secondary disease reactivated bacterial infections of various tissues may then complicate the organic lesions caused by the GVH-reaction, resulting in the high mortality rate of "conventional" allogeneic chimeras.

The comparable chance of survival of allogeneic decontaminated and isogeneic bone marrow chimeras goes hand in hand with a very similar pattern of hemopoietic recovery, when the difference in the repopulating ability under the two experimental conditions is taken into account [4].

By means of hemoglobin electrophoresis, hemopoietic chimerism was proven to be established as far as the erythrocytic differentiation of the stem cells transplanted is concerned. Although no experimental data have been presented to identify the genetic origin of all hemopoietic cell lines, it is most likely, that both, granulopoiesis and thrombopoiesis were of donor cell origin, too. For the lymphopoietic system Huget et al. [5] have shown that the immunocompetent cells of all allogeneic chimeras under investigation were of $C_{57}$Bl specificity when assayed in mixed lymphocyte cultures and by the cytoxicity test.

As shown by a preliminary experiment (Table 2), bone marrow cells of allogeneic chimeras seem to be effective to protect lethally irradiated "isogeneic" and "allogeneic" mice. The addition of spleen cells as a source of immunocompetent cells, however, failed to induce GVH-reaction in any of the recipients. From these data, we may tentatively conclude, that in both, the bone marrow and the spleen of long-term chimeras, the immunocompetent cells may have lost their reactivity on the same type of alloantigens confronted with in the primary host. Whether the nature of tolerance observed in this experiment may be understood as the result of an impaired recovery of the lymphatic tissue or as a selective regeneration of subpopulations of lymphocytes (e.g. T-cells) remains open. Recent data of Bach et al. [1] and Sprent et al. [10] provided experimental evidence of two different T-cell specificities involved in alloantigenic reactions. One population of T-lymphocytes being selectively responsive to "lymphocyte defined" histoantigens, whereas the others would cover the alloantigens different from determinants of lymphocytic cells, are eradicated by lethal irradiation, and, furthermore, the time of the onset of GVH-reaction and the mortality rate of allogeneicly grafted hosts is understood to be a function of the number of T-cells grafted, we may deduce, that very few T-cell numbers, ready to confront other than lymphohemopoietic cell bound alloantigens are available in the long term chimera, thus explaning the failure to induce secondary disease in the second conventional recipient.

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